

Blind Audit of J-Lens / R-Lens / Logit-Lens on Qwen3.6-27B

Executive Summary (DRAFT - agent synthesis, provisional). Final exec summary & pivot are the human's.

1. Question and hypotheses

Does a lens's signal in a blind scan survive when the text-reconstruction channel is neutralized?

Three competing hypotheses are discriminated:

- H1 - internal readout: the lens surfaces a genuine internal signal, beyond surface text.
- H2 - text inversion: the apparent signal is just the scan letting a reader reconstruct the text.
- H3 - judge apophenia: the judge manufactures 'anomalies' from anomaly-shaped noise.

2. Method

11 human-validated anomalous/clean text pairs (families: bug, false_premise, injection). Each pair is scanned by 3 instruments (J-lens, R-lens, logit lens) into per-position, per-layer top-k token stacks. Two LLM judges (judgeA = claude-sonnet-4.6; judgeB = gpt-4.1-mini) score each scan for anomaly + confidence, blind to any anomaly family or instrument name. Seven fixed conditions include the controls that carry the argument: prompt-only (text baseline), blind scan, scan_permuted (positions shuffled = H3 control), and reconstruction (judge A rebuilds text from the scan, judge B reads only that = H2 control). A leak audit gates every scan run.

3. Headline results (all figures/numbers computed this session; see Result.md)

Leak gate GREEN: 0 shared 4-grams across 66 scans; no text-reconstruction leak. 579/592 valid verdicts (2.2% loss). Independent from-scratch AUC recompute coincides with the pipeline to $<1e-6$.

- Text baseline is near-ceiling: prompt-only AUC 0.95 (judgeA) / 0.96 (judgeB).
- Pooled blind-scan AUC per instrument: logit 0.84 $\geq$ rlens 0.80 $\geq$ jlens 0.75 (n=11/11).
- Triplet (dAUC, 95% paired-bootstrap CI), pooled over judges/prompts:
 - D1 scan - prompt: jlens -0.21 [-0.47,0.01] | rlens -0.16 [-0.30,-0.02] | logit -0.12 [-0.30,0.05]
 - D2 scan - reconstruction: jlens +0.11 [-0.30,0.47] | rlens +0.49 [0.23,0.79] | logit +0.32 [0.05,0.57]
 - D3 permuted - chance: jlens +0.28 [0.02,0.46] | rlens +0.20 [0.01,0.39] | logit +0.29 [0.11,0.45]
- Inter-judge agreement (scan, v1): Cohen kappa 0.49, Spearman 0.50 (moderate).
- Evidence fidelity: ~52% of tokens the judge CITES are absent from the scan; ~13% of the present ones fall in the anomaly zone. High false-alarm on CLEAN twins (reconstruction 0.5-1.0).

4. What the data asserts (factual)

- D3 > 0 with CI excluding 0 on all three instruments: shuffling positions does NOT destroy the discriminative signal -> a large part of it is order-invariant.
- D1 < 0 everywhere: the blind scan never beats reading the text.
- The logit lens - which has no privileged internal-readout machinery - gives the HIGHEST blind-scan AUC, matching/beating the two workspace lenses.
- The judge fabricates the majority of its cited evidence and flags clean twins often.

5. Bugs and anomalies encountered (and fixed)

- judgeB (gpt-5-mini) reasoning-truncated EVERY JSON verdict (finish_reason=length, empty) -> switched to gpt-4.1-mini (non-reasoning, still OpenAI family).
- Judge client had no timeout -> a stalled call hung the whole run; max_tokens=600 too low for claude on verbose scans -> added timeout=90 and raised to 1200.
- content_filter: claude refuses to read some reconstructed texts, biased onto false_premise (~10 lost keys) -> makes D2 fragile.
- Windows cp1252 vs UTF-8 crash on every scan read -> fixed at the root (encoding='utf-8').

6. Limitations / caveats

- $n = 11$ pairs: pooled CIs are wide; per-family AUCs are exploratory noise.
- The permutation control preserves token CONTENT (only order is shuffled), so $D3 > 0$ cannot separate apophenia (H3) from a real-but-position-invariant readout. This is the key design gap.
- D2 compares the scan to a BROKEN reconstructor (high false-alarm, degenerate outputs, content_filter hole) -> it cannot cleanly reject H2.
- Judge B was changed mid-project (gpt-5-mini -> gpt-4.1-mini); results differ from the frozen plan.

7. Provisional conclusion (AGENT OPINION - to be verified, not the pivot)

The available evidence LEANS AGAINST H1 (privileged internal readout): the 'dumb' logit lens leads on the blind scan, and the signal is order-invariant. H2 vs H3 stays UNRESOLVED because the reconstruction control is broken and the permutation control does not destroy token content. Honest headline: no clean survival of a lens-specific signal once surface vocabulary is controlled, at $n=11$. This matches the project's own prior: the value is in the controls, not the raw AUC.

8. Potential leads / next steps

- Add a FOREIGN-SCAN control (judge a different clean item's scan) to destroy token content and actually separate H3 from a position-invariant readout - cheapest high-value experiment.
- Fix or drop the reconstruction channel before trusting D2 (content_filter + degeneracy).
- Classify ALL false positives (not just the 5 in the seed-0 sample) into (a) evokes family / (b) evokes own content / (c) invents; proportion (c) is the direct H3 estimate.
- Scale n beyond 11 if budget allows; report only pooled AUC + triplet as primary.

9. Budget and artifacts

Spend 14.97/19. Artifacts:

results/{metrics,triplet,inter_judge,leak_check,evidence_check,family_false_alarm,blockH_worksheet},
figs/fig1_triplet.png, fig2_auc_family.png, judge/outputs/verdicts.jsonl, Result.md (full record),
experiments.md entry (14).

Figure 1 - H1/H2/H3 triplet per instrument (95% paired-bootstrap CI)

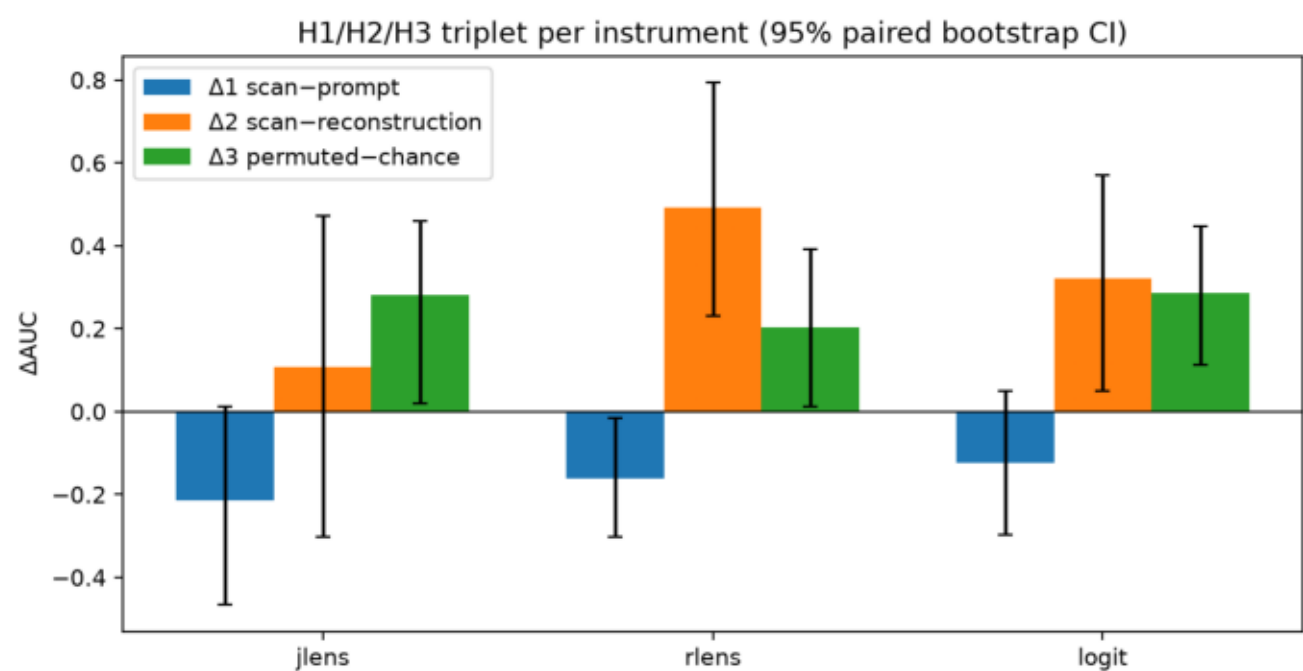


Figure 2 - AUC per family x instrument (exploratory, n~4/family)

